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Gondo et al.

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(54) **SERVER DEVICE, INFORMATION PROCESSING METHOD, AND COMPUTER PROGRAM PRODUCT**

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CPC **H04N 21/23106** (2013.01); **H04L 65/61**
(2022.05); **H04N 21/2393** (2013.01)

(58) **Field of Classification Search**
CPC H04N 21/23106; H04N 21/2393; H04L
65/61

See application file for complete search history.

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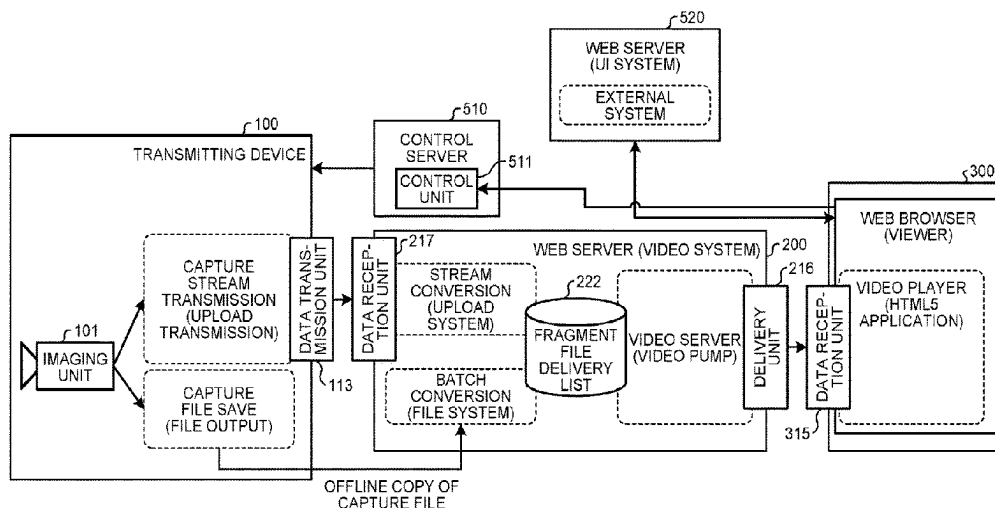
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(57) **ABSTRACT**

According to an embodiment, a server device includes one or more processors and a first storage. The processors receive first data of transmission data from a transmitting device. The processors further receive second data of the transmission data, transmitted from the transmitting device in response to a transmission request. The processors deliver the first data and the second data to a receiving device. The first storage stores the first data and the second data. The processors delete at least one of the first data and the second data from the first storage according to a predetermined condition, and store data including at least a portion of the deleted data in a second storage.

14 Claims, 14 Drawing Sheets



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FIG. 1

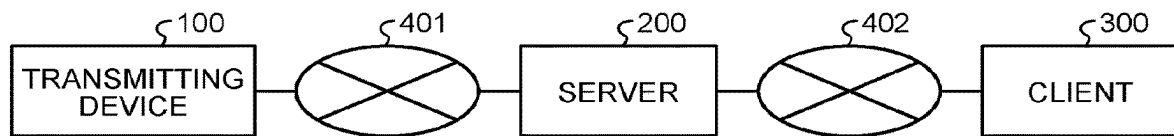


FIG. 2

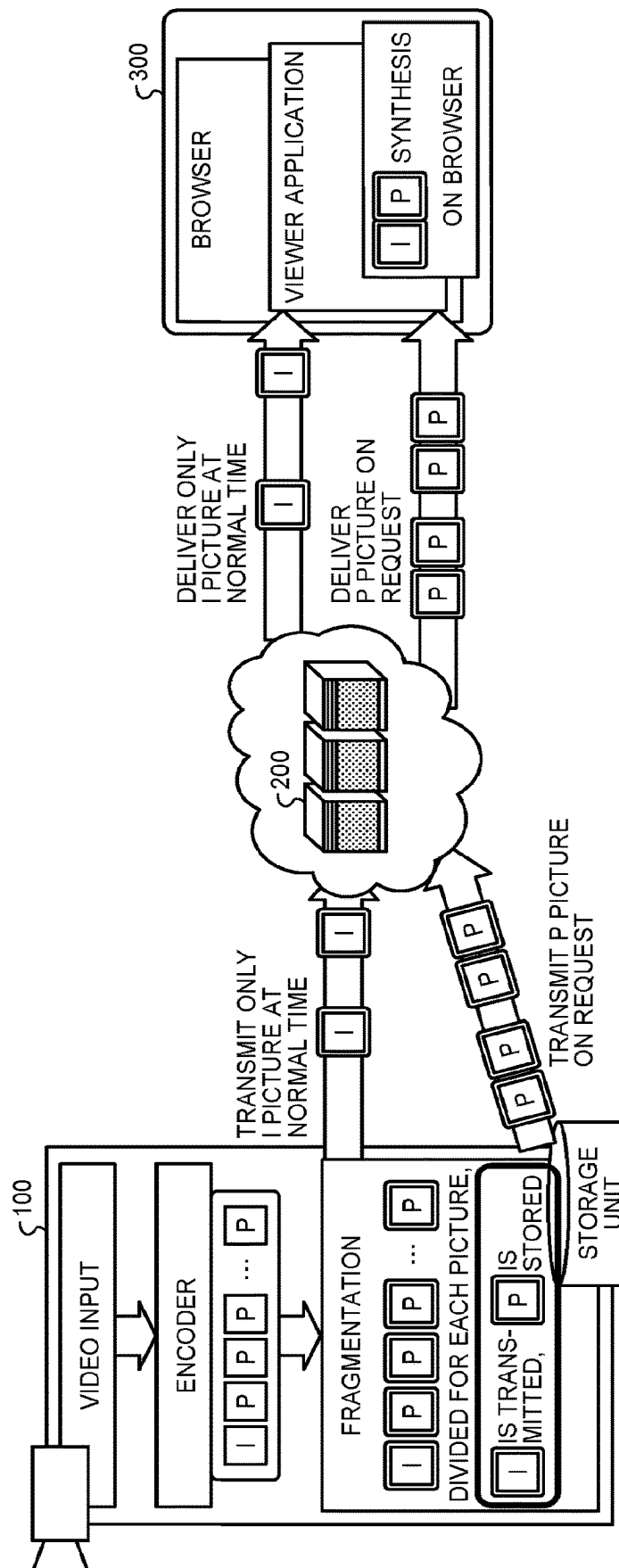


FIG. 3

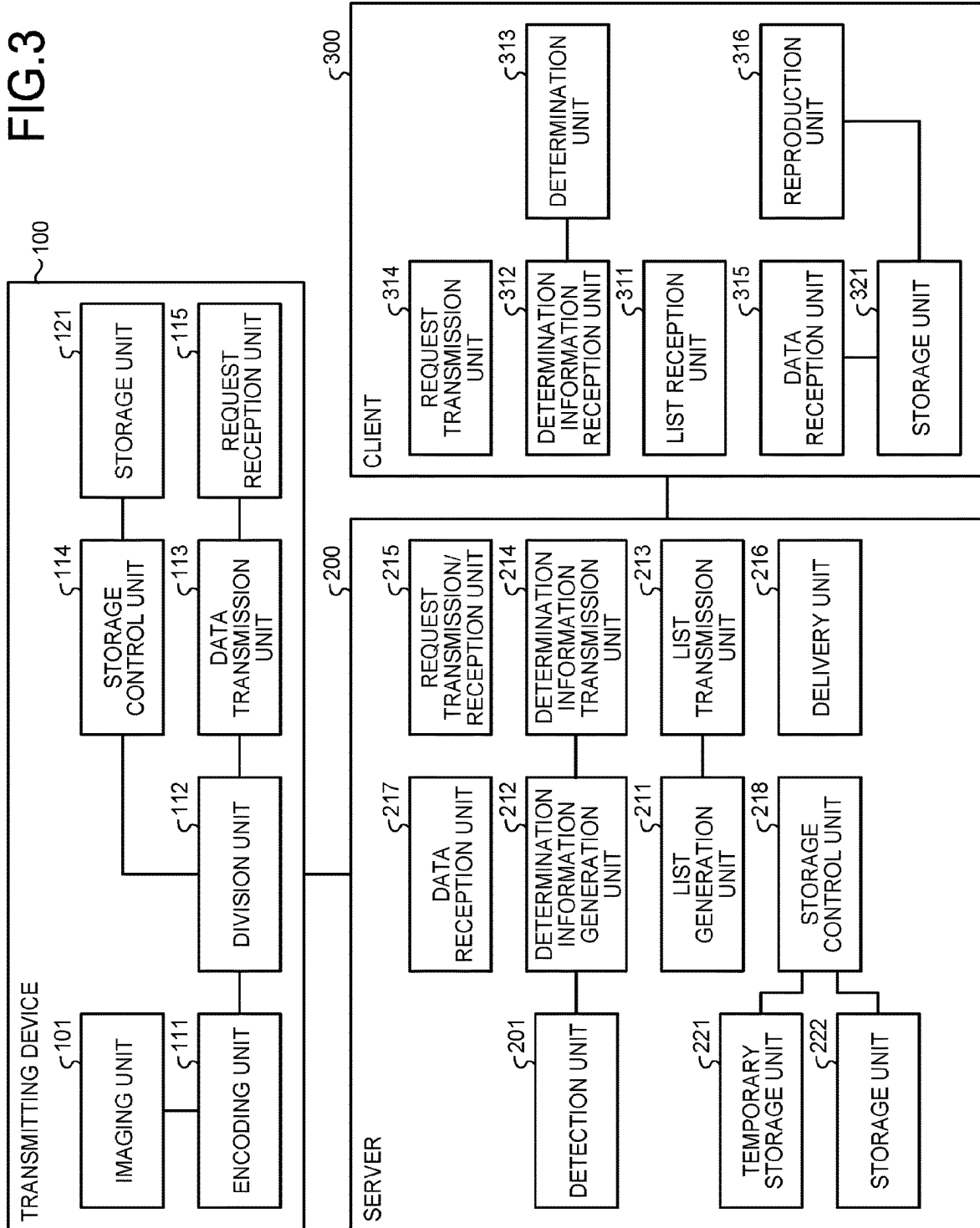


FIG.4

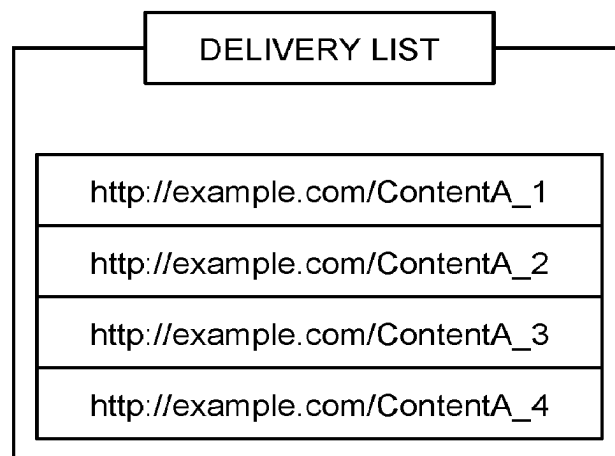


FIG.5

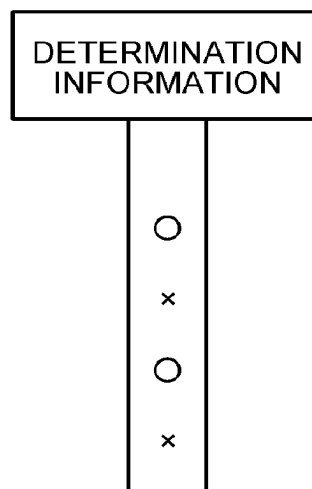


FIG.6

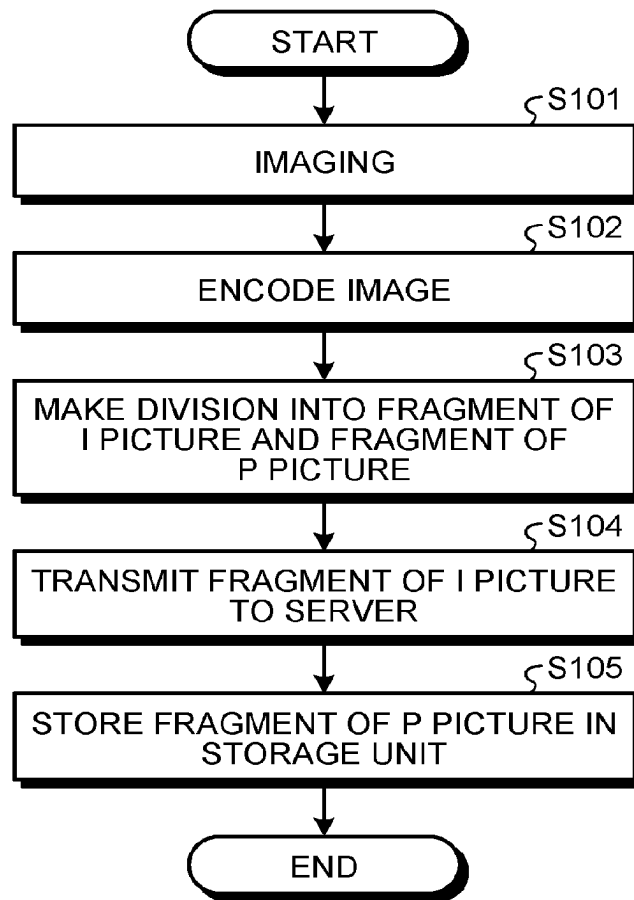


FIG.7

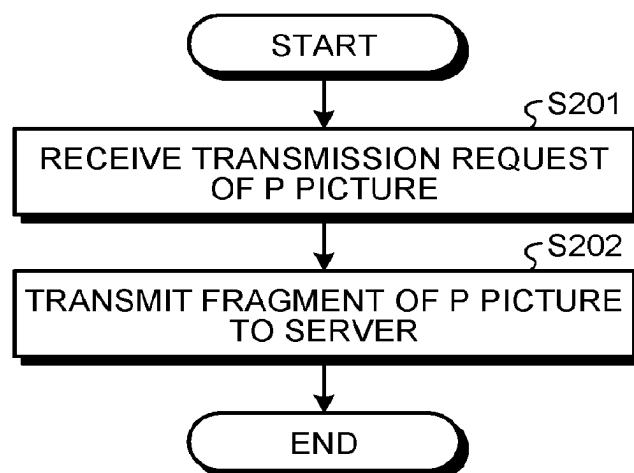


FIG.8

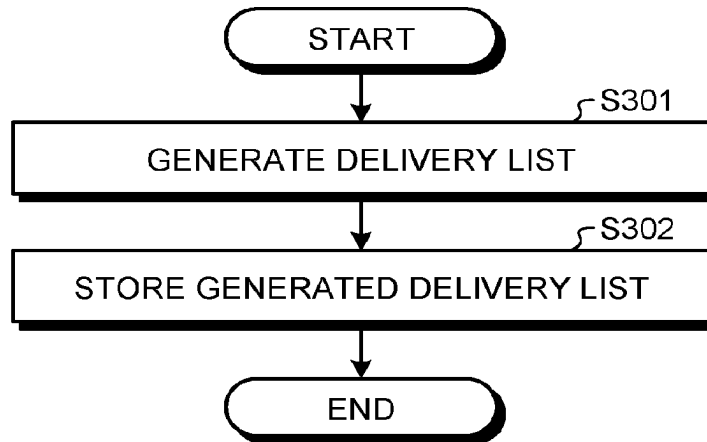


FIG.9

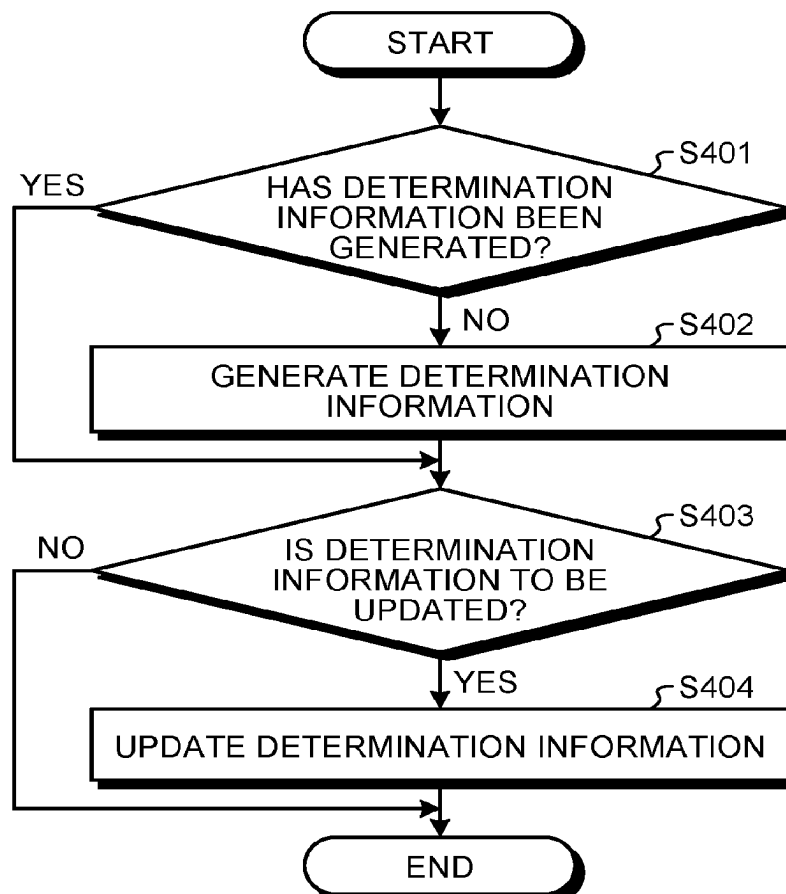


FIG.10

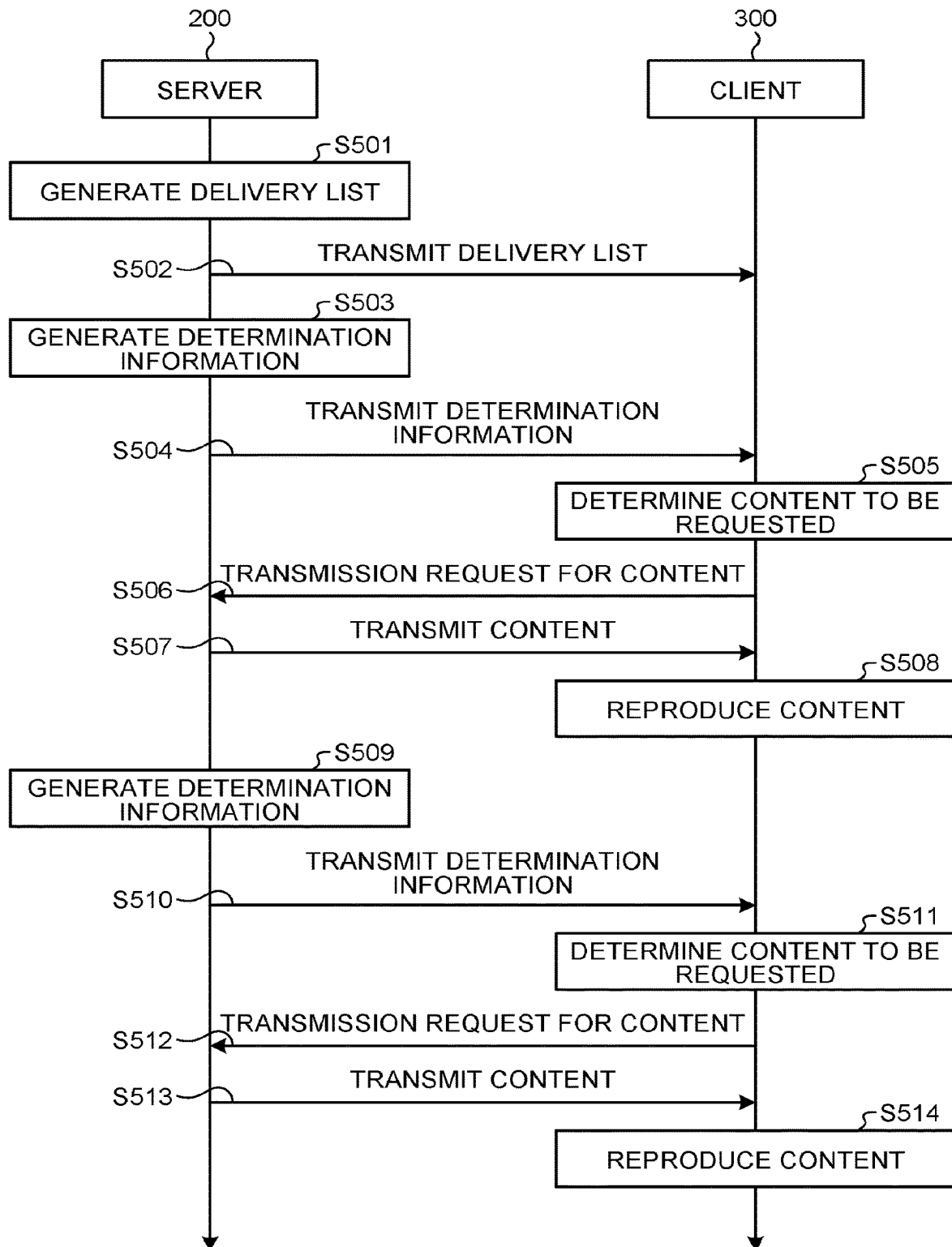


FIG.11

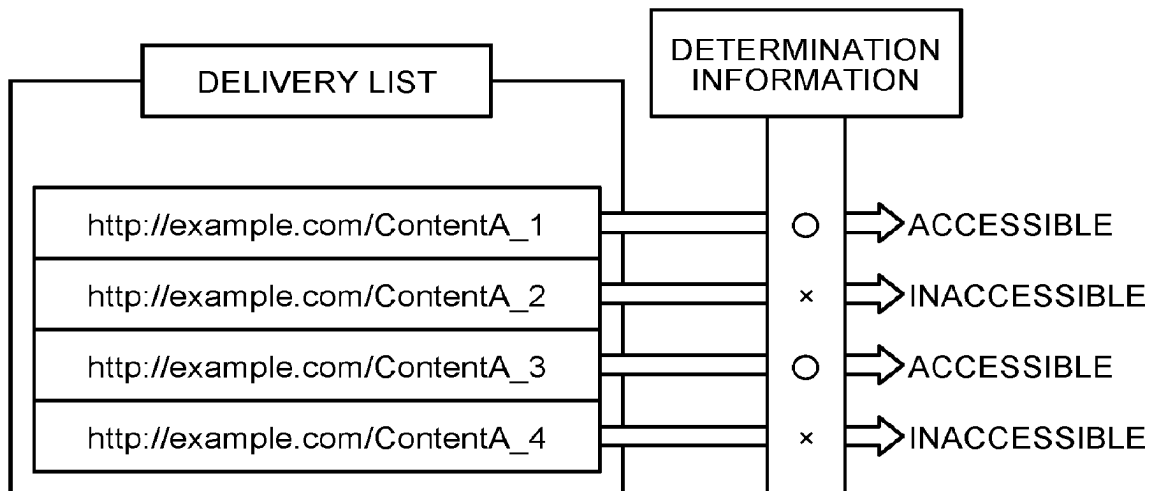


FIG.12

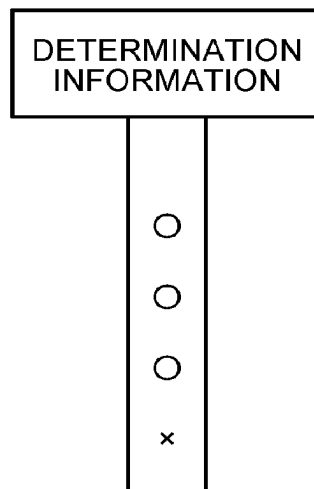


FIG.13

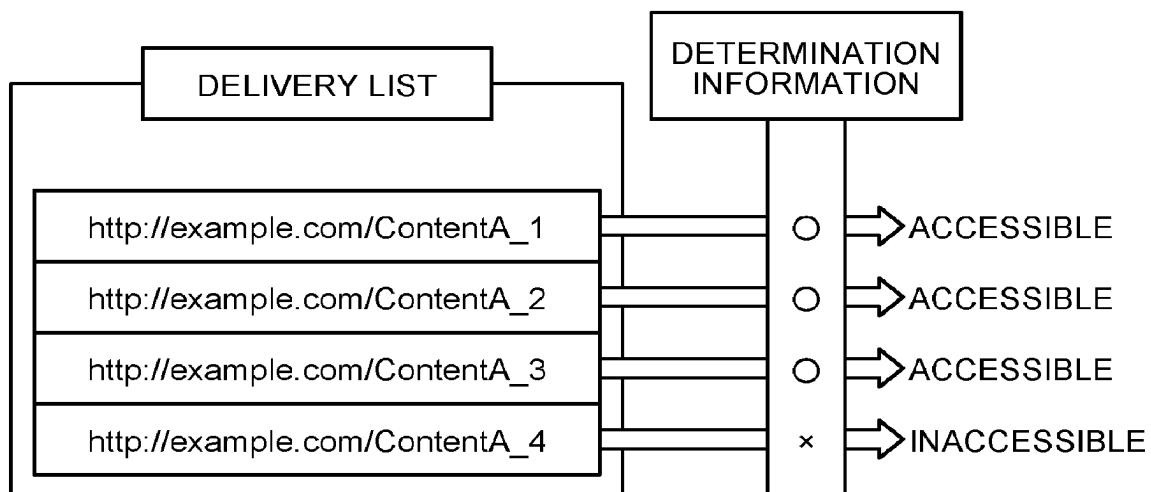


FIG.14

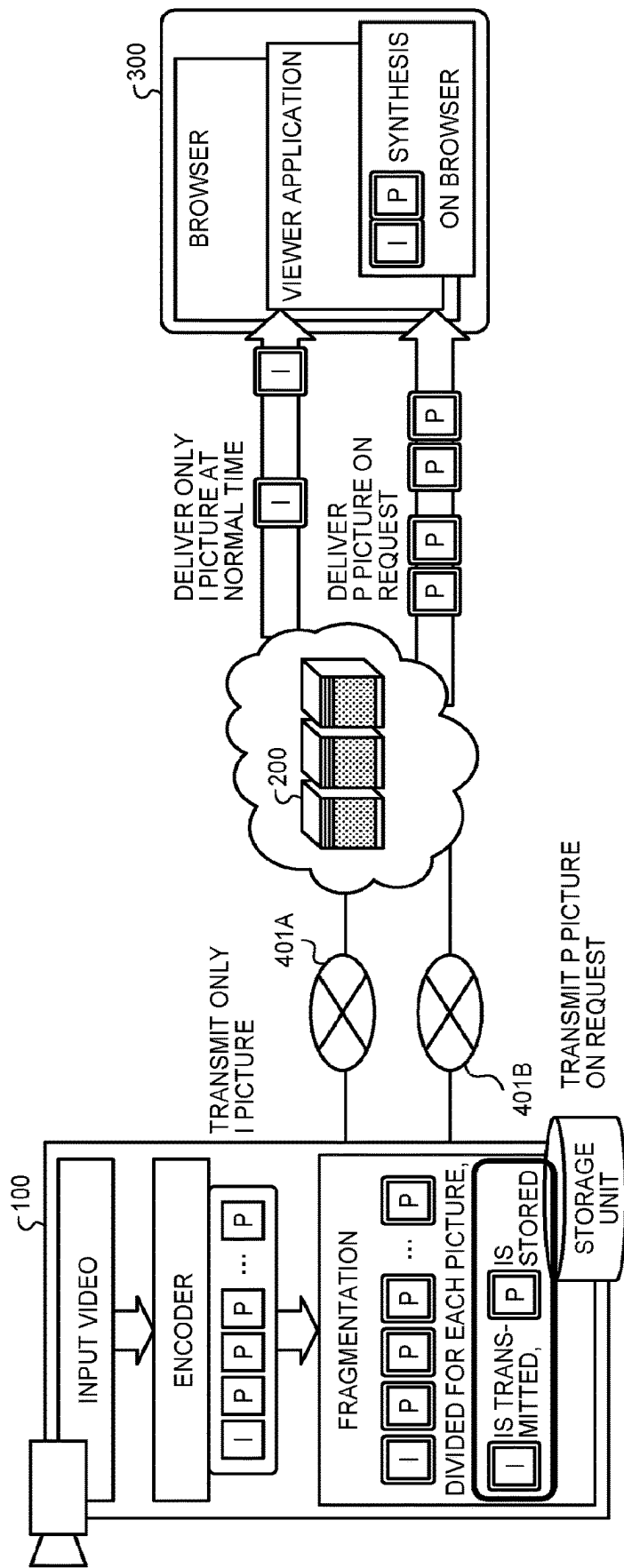


FIG.15

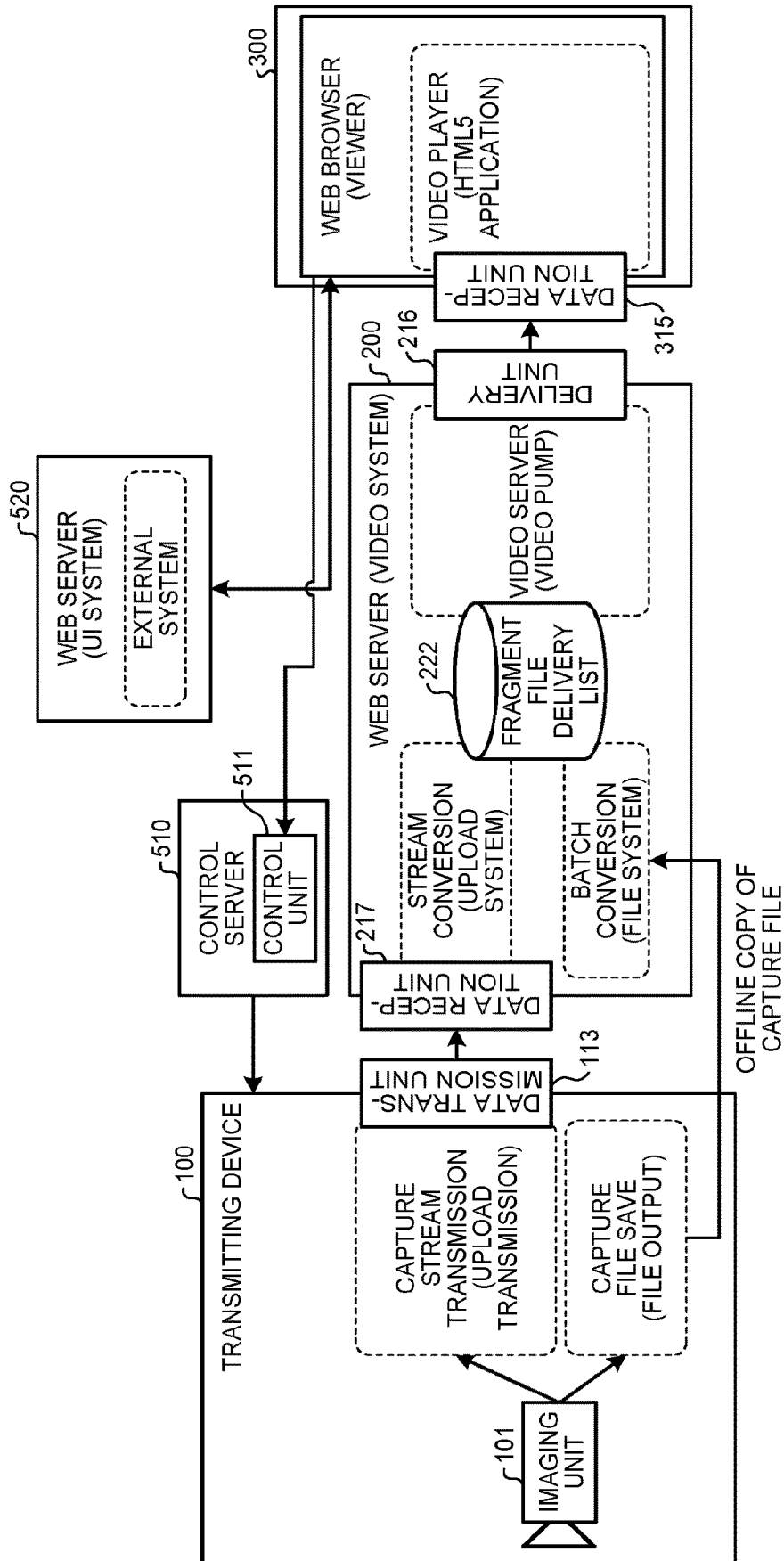


FIG. 16

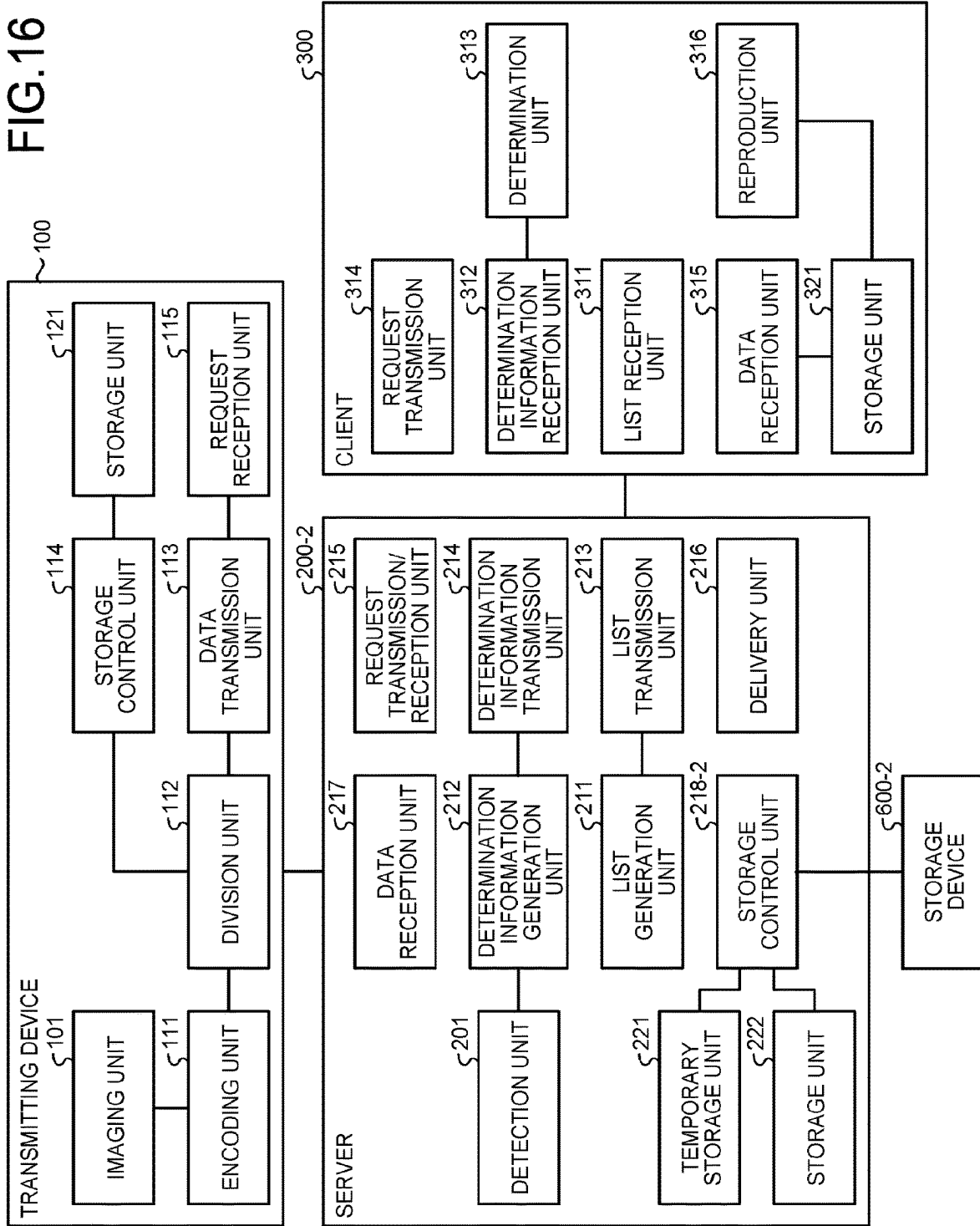


FIG.17

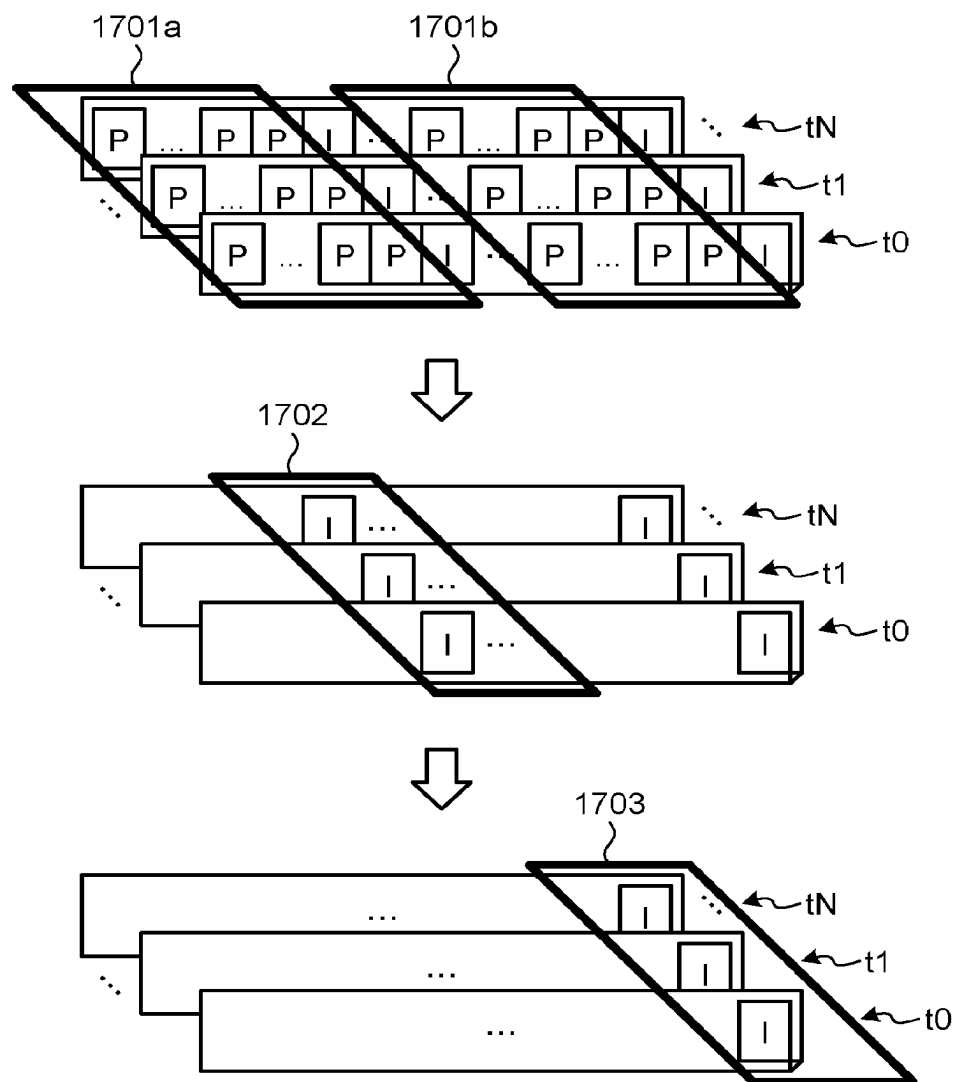
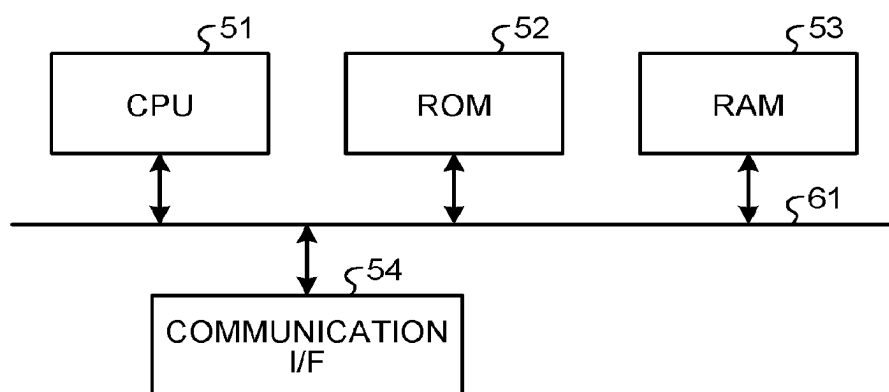


FIG.18



1

SERVER DEVICE, INFORMATION PROCESSING METHOD, AND COMPUTER PROGRAM PRODUCT

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/185,491, filed Feb. 25, 2021, which is based upon and claims the benefit of priority from Japanese Patent Application No. 2020-075375, filed on Apr. 21, 2020; the entire contents of which are incorporated herein by reference.

FIELD

Embodiments described herein relate generally to a server device, an information processing method, and a computer program product.

BACKGROUND

Adaptive streaming such as HTTP Live Streaming (HLS) and Moving Picture Experts Group-Dynamic Adaptive Streaming over HTTP (MPEG-DASH) can be applied to, for example, a system (a video delivery system and a video monitoring system) that delivers and monitors video (moving image data) captured by a camera.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram of a video delivery system according to the present embodiment;

FIG. 2 is a diagram illustrating an outline of delivery processing by the video delivery system of the present embodiment;

FIG. 3 is a functional block diagram of each device of the video delivery system of the present embodiment;

FIG. 4 is a diagram illustrating an example of a data structure of a delivery list;

FIG. 5 is a diagram illustrating an example of a data structure of determination information;

FIG. 6 is a flowchart of transmission processing according to the present embodiment;

FIG. 7 is a flowchart of transmission processing of stored data;

FIG. 8 is a flowchart of list generation processing according to the present embodiment;

FIG. 9 is a flowchart of determination information generation processing according to the present embodiment;

FIG. 10 is a sequence diagram of content delivery processing in the present embodiment;

FIG. 11 is a diagram illustrating an example of determining a delivery list using the determination information;

FIG. 12 is a diagram illustrating an example of updated determination information;

FIG. 13 is a diagram illustrating an example of determining a delivery list using the determination information;

FIG. 14 is a diagram illustrating a configuration example of a video delivery system of a modified example;

FIG. 15 is a diagram illustrating an outline of delivery processing by the video delivery system of the modified example;

FIG. 16 is a block diagram of the video delivery system of the modified example;

FIG. 17 is a diagram illustrating an example of how to move data; and

2

FIG. 18 is a hardware configuration diagram of a transmitting device according to the present embodiment.

DETAILED DESCRIPTION

According to an embodiment, a server device includes one or more processors and a first storage. The processors receive first data of transmission data from a transmitting device. The processors further receive second data of the transmission data, transmitted from the transmitting device in response to a transmission request. The processors deliver the first data and the second data to a receiving device. The first storage stores the first data and the second data. The processors delete at least one of the first data and the second data from the first storage according to a predetermined condition, and store data including at least a portion of the deleted data in a second storage.

A preferred embodiment of a transmitting device according to the present invention will be described in detail below with reference to the accompanying drawings.

In a conventional video delivery system using adaptive streaming, there is a need to prepare a network line with a high transmission rate in order to reliably record a high-quality video stream (an example of transmission data) required for high-quality live video delivery.

On the other hand, in a large-scale video delivery system in which a large number of cameras are connected, a large amount of camera video is always transmitted. Therefore, in order to reduce a line load or cost, it is desirable to use a network line having a low transmission rate between the camera and a server device for recording video. However, on a line with a low transmission rate, a quality of live video will be low, which may interfere with monitoring work.

Therefore, in the present embodiment, a video delivery system capable of reducing a line load or a processing load without deteriorating a quality is realized.

Here, an example of transmission data to which the present embodiment can be applied will be described. The transmission data includes time series data in which a time order is defined, such as moving image data and sensor data.

The moving image data is, for example, data captured by an imaging device such as a camera and a frame capture. The moving image data is acquired in real time from, for example, the imaging device, and is used as content to be delivered. The moving image data once stored in a storage medium after being captured may be used as the content to be delivered.

The sensor data is data indicating a value detected by a sensor (detection device), and includes, for example, information on the time when the data is detected (sampled). The sensor may be any device. For example, a microphone that acquires voice, a Global Positioning System (GPS) device that acquires location information, and a sensor that periodically or irregularly detects a surrounding environment to be detected, the temperature, speed, pressure, and the like of an electronic device and outputs it as sensor data can be used.

In the following, an example in which the moving image data is mainly used as transmission data will be described.

FIG. 1 is a block diagram illustrating an example of a configuration of the video delivery system according to the present embodiment. As illustrated in FIG. 1, the video delivery system includes a transmitting device 100 (an example of a transmitting device), a server 200 (an example of a server device), and a client 300 (an example of a receiving device). The transmitting device 100 and the

3

server **200** are connected by a network **401**. The server **200** and the client **300** are connected by a network **402**.

The networks **401** and **402** can be any network, such as the Internet. For example, the networks **401** and **402** may be either a wired network or a wireless network. In addition, the networks **401** and **402** may be configured as one integrated network.

A configuration of the communication system illustrated in FIG. **1** is an example, and is not limited to thereto. For example, a plurality of transmitting devices **100**, servers **200**, and clients **300** may be provided. In addition, each of the transmitting device **100**, the server **200**, and the client **300** may be physically configured by one device, or may be physically configured by a plurality of devices. For example, the server **200** may be built on a cloud environment.

FIG. **2** is a diagram illustrating an outline of delivery processing by the video delivery system of the present embodiment. The transmitting device **100** inputs a video and encodes (codes) the input video. The transmitting device **100** compresses and encodes the video according to a standard such as H.264. The encoded video includes, for example, I pictures and P pictures. The I picture is image data encoded by intra-frame prediction, and includes a full-screen range that can be reproduced independently. The P picture is image data encoded by inter-frame prediction based on the I picture. The P picture cannot be reproduced alone, but can be reproduced in combination with the I picture.

The transmitting device **100** divides (fragments) the encoded images into pictures. Then, the transmitting device **100** transmits a portion of the divided pictures (for example, I picture) to the server **200**, and stores the remainder (for example, P picture) in a storage unit. The transmitting device **100** may also store the picture transmitted to the server **200** in the storage unit for playback (local reproduction), backup, and the like.

Normally, the server **200** delivers only the I picture transmitted from the transmitting device **100** to the client **300**. The client **300** displays, for example, a video delivered using a viewer application. The viewer application is, for example, an application included in a browser for displaying and viewing a video. Since the I pictures are transmitted at regular intervals (for example, several images per second or one image every few seconds), a video (para-para cartoon, frame-by-frame (intermittent) video) in which a still image (I picture) is updated at regular intervals is displayed on the client **300**. The viewer application can be realized as an application using Media Source Extensions (MSE), which is an application programming interface for Hyper Text Markup Language (HTML) **5**, for example. This enables streaming reproduction using HTTP download.

When the display of the P picture is requested by a user who operates the client **300** or the like, the transmitting device **100** reads the stored P picture and transmits the P picture to the server **200**. The server **200** delivers the transmitted P picture to the client **300**. The server **200** may deliver the transmitted P picture as it is, or deliver the P picture to which only processing (such as changing a transmission packet format) other than conversion processing (such as recompression) for image data is applied. The viewer application of the client **300** synthesizes and displays the I picture already received and the P picture received later. As a result, the client **300** can display a smoother image. If the transferred data (for example, I picture) is stored in the storage unit in the client **300**, it can be used for synthesizing with the P picture received later, and the like. That is, it is not necessary to transmit the transferred data from the transmitting device **100** and the server **200** again. Therefore,

4

the amount of transmission data can be reduced as compared with the method of retransmitting all the video data including the P picture.

As described above, the server **200** does not apply conversion processing such as recompression to the image data, and delivers the image data transmitted from the transmitting device **100** to the client **300** as it is or by processing only the transmission packet format. For example, the server **200** does not need to perform processing of converting to a video with reduced quality in order to enable delivery at a low transmission rate. Therefore, it is possible to avoid an increase in the processing load of the server **200** and deterioration of image quality. In addition, in the normal state, since only the I picture is delivered, the amount of communication can be suppressed. In this way, it is possible to reduce the line load and the processing load without deteriorating the quality.

Note that the I picture is a portion of the data constituting the video (moving image data), but can be treated as a still image. For example, the browser (viewer application) running on the client **300** may be able to display the I picture included in the moving image data as still image data. Therefore, the I picture can be displayed as the still image data without performing compression processing on a still image such as Joint Photographic Experts Group (JPEG). Compared to JPEG format still images, since the I picture has a smaller data size (higher compression efficiency), the line load and the processing load can be further reduced. In addition, even when the delivered video is used for image recognition, the I picture can be used as input data for image recognition without converting the video into the still image.

Next, the details of the configuration of each device of the video delivery system of the present embodiment will be described. FIG. **3** is a block diagram illustrating an example of a functional configuration of each device of the video delivery system of the present embodiment.

As illustrated in FIG. **3**, the transmitting device **100** includes an imaging unit **101**, a storage unit **121**, an encoding unit **111**, a division unit **112**, a data transmission unit **113**, a storage control unit **114**, and a request reception unit **115**.

The imaging unit **101** captures and outputs a video (moving image data). The imaging unit **101** can be realized by, for example, an imaging element such as a Charge Coupled Device (CCD) and a CMOS image sensor (CIS), or a frame buffer capture such as a frame memory, a frame grabber, and a screen capture.

The storage unit **121** stores various data used in various processing by the transmitting device **100**. For example, the storage unit **121** stores the video captured by the imaging unit **101**.

The encoding unit **111** encodes the video input from the imaging unit **101**. An encoding method by the encoding unit **111** may be any method, and for example, an encoding method according to a standard such as H.264 can be applied. The encoding unit **111** compresses and encodes the video, for example, and outputs a video including an I picture and a P picture. Each picture of the encoded video corresponds to a plurality of transmission data to be transmitted.

The division unit **112** divides the encoded video into data (first data) to be transmitted to the server **200** and data (second data) to be stored in the storage unit **121** without transmission. For example, the division unit **112** divides the encoded image for each picture so that the I picture is data (fragment data) to be transmitted to the server **200** and the P picture is data (fragment data) to be stored in the storage unit **121**. The division unit **112** may divide the video so that

5

each of the divided pictures becomes one file (fragment file). Regarding the data stored in the storage unit 121, the division unit 112 may divide the data so that a plurality of pictures are included in one file.

The method of dividing the data by the division unit 112 is not limited thereto, and any method may be used. For example, the division unit 112 may divide the encoded video into an I picture selected from a plurality of I pictures at a fixed number and other pictures (remaining I pictures and P pictures). In addition, for example, the division unit 112 may divide the encoded video into a P picture selected from the I picture and a plurality of P pictures at a fixed number and other pictures (remaining P pictures). In addition, for example, the division unit 112 may divide the encoded video into an I picture selected from a plurality of I pictures at a fixed number, a P picture selected from a plurality of P pictures at a fixed number, and other pictures (remaining I pictures and P pictures).

The data transmitted to the server 200 may be represented in a format that the server 200 can deliver as it is. For example, the division unit 112 may convert the divided data so that the format conforms to a standard such as Fragmented MP4 (fMP4).

The data transmitted to the server 200 may be represented in a format to which data (metadata) that can be converted into a format delivered by the server 200 is added. For example, the division unit 112 may add metadata including information necessary for conversion to the divided data so that the server 200 can convert the data into a format according to a standard such as fMP4. The information necessary for conversion includes, for example, information (IP address, port number, etc.) that identifies the transmitting device 100, time (such as the date, time, minute, and second of the image), and a position of the image within the time (for example, information indicating the number from the beginning).

The division unit 112 may change the size or encode amount of the data transmitted to the server 200 according to a band of the network 401. For example, the division unit 112 may divide data corresponding to a size that can be transmitted within an allocated band of the network 401 without delay or the like, or data having an encoding bit rate within the band of the network 401 as data to be transmitted to the server 200. For example, in the case where the band of the network 401 is large, the division unit 112 divides a portion of the I picture and the P picture as the data to be transmitted to the server 200, and may divide only the I picture or a portion of the I picture (such as thinning out the I pictures at a fixed number) as the data to be transmitted to the server 200 as the band becomes smaller. As an encoding method for partially delivering the P picture, for example, a method described in Japanese Patent No. 6239472 can be used.

The division unit 112 may change the size or encode amount of the data transmitted to the server 200 by dynamically switching the division method as described above according to the band of the network 401 or the request from the client 300 or the server 200.

If the data encoded by the encoding unit 111 is simply divided, the size of the divided data may be inconsistent. For example, in a case where the encoding unit 111 has a function of adjusting the encode amount of each picture according to the band of the network 401, the size of each I picture and the size of each P picture may be different from each other. Therefore, among each of the pictures encoded in this way, for example, if the I picture is divided as the data

6

to be transmitted to the server 200, the size of each of the divided I pictures may also be different from each other.

Therefore, the encoding unit 111 may adjust the encode amount according to the band of the network 401 so that the size or the encoding bit rate of the divided data becomes the size or the encoding bit rate that can be transmitted within the band of the allocated network 401 without delay or the like. For example, if the division unit 112 divides the video so that only the I picture is transmitted to the server 200, the encoding unit 111 may encode the video so that the size of the I picture becomes a certain size that can be transmitted within the band of the network 401 without delay or the like, or becomes the encoding bit rate within the band of the network 401.

The data transmission unit 113 transmits data to an external device such as the server 200. For example, the data transmission unit 113 transmits the data (first data) to be transmitted to the server 200 among the data divided by the division unit 112 to the server 200. In addition, in a case where the data transmission unit 113 receives a data transmission request stored in the storage unit 121 by the request reception unit 115 (described later), the data transmission unit 113 transmits requested data to the server 200.

The storage control unit 114 controls storage processing for the storage unit 121. For example, the storage control unit 114 stores the data that is not transmitted to the server 200 among the data divided by the division unit 112 in the storage unit 121. The storage control unit 114 may store the data (first data) transmitted to the server 200 in the storage unit 121. At this time, the storage control unit 114 may manage the stored data using metadata indicating that the transmitted data (first data) has been transmitted or that the second data has not been transmitted. The storage control unit 114 may delete the data stored in the storage unit 121 according to a predetermined condition. For example, the storage control unit 114 may delete the data for which a certain period of time has passed since the data was stored.

The request reception unit 115 receives a transmission request of the data stored in the storage unit 121 from the server 200. In the case of a configuration in which the request is transmitted from the client 300 via a control server or the like other than the server 200, the request reception unit 115 may receive the transmission request for data stored in the storage unit 121 from such a control server.

Each of the above units (the encoding unit 111, the division unit 112, the data transmission unit 113, the storage control unit 114, and the request reception unit 115) is realized by, for example, one or more processors. For example, each of the units may be realized by causing a processor such as a central processing unit (CPU) to execute a program, that is, by software. Each of the units may be realized by a processor such as a dedicated integrated circuit (IC), that is, hardware. Each of the units may be realized by using software and hardware in combination. When using a plurality of processors, each processor may realize one of the units or two or more of the units.

Note that each function of the transmitting device 100 may be delivered to a plurality of physically or logically different devices. For example, it may be divided into a device (video input and encoder in FIG. 2) including the imaging unit 101 and the encoding unit 111 and a device (fragmentation and storage unit in FIG. 2) including each remaining unit. In this case, the data encoded by the encoding unit 111 is input to the latter device by, for example, a communication path such as a network or a coaxial cable. The latter device may be configured to acquire the encoded data via the communication path.

Next, a configuration of the server **200** will be described. The server **200** delivers a delivery list and determination information to the client **300** via the network **402**. The delivery list is a list that describes information about the data to be delivered (hereinafter, also referred to as content). Usually, a content deliverer describes metadata such as a content acquisition destination and a bit rate in the delivery list. A content acquirer can specify the content to be acquired by acquiring and analyzing the delivery list.

The content described in the delivery list may include not only content that can be transmitted, but also content that cannot be transmitted, content that is not permitted to be transmitted, and the like. The determination information is information for the receiving device (client **300**) to determine whether or not to request transmission of the content included in the delivery list.

As illustrated in FIG. 3, the server **200** includes a detection unit **201**, a list generation unit **211**, a determination information generation unit **212**, a list transmission unit **213**, a determination information transmission unit **214**, a request transmission/reception unit **215**, a delivery unit **216**, and a data reception unit **217**, a storage control unit **218**, a temporary storage unit **221**, and a storage unit **222**.

The detection unit **201** detects that the content can be transmitted. For example, the detection unit **201** determines that the content can be transmitted when the content is provided by a providing device (such as the transmitting device **100**) that provides the content. The detection unit **201** may monitor a storage area (for example, the temporary storage unit **221**) in which the content is stored, and may determine that the content can be transmitted when the content is stored.

The list generation unit **211** generates a delivery list. The list generation unit **211** generates the delivery list, for example, when the client **300** requests to generate and transmit the delivery list. A trigger for generating the delivery list is not limited thereto, and may be any trigger. For example, the list generation unit **211** may generate the delivery list of contents to be transmitted in a next period every time a certain period of time elapses. The list generation unit **211** may generate the delivery list when the content is provided by the providing device or when the generation is instructed.

FIG. 4 is a diagram illustrating an example of a data structure of the delivery list. As illustrated in FIG. 4, the delivery list includes identification information that identifies the content. FIG. 4 illustrates an example of using a Uniform Resource Locator (URL) of the content as the identification information. As the identification information, information other than the URL may be used as long as the content can be identified. The delivery list may include information other than the identification information. In the present embodiment, for example, the delivery list including the URL for each divided data (picture, etc.) is created.

Returning to FIG. 3, the determination information generation unit **212** generates determination information. For example, the determination information generation unit **212** generates determination information indicating that the content can be transmitted when the content can be transmitted, and generates determination information indicating that the content cannot be transmitted when the content cannot be transmitted. In addition, when a situation indicating whether or not the content can be transmitted changes, the determination information generation unit **212** generates updated determination information to correspond to the changed situation.

The determination information generation unit **212** generates the determination information, for example, when the client **300** requests that the determination information is generated and transmitted. When the delivery list is generated by the list generation unit **211**, the determination information generation unit **212** may generate the determination information at that time. A trigger for generating the determination information is not limited thereto, and may be any trigger. For example, the determination information generation unit **212** may detect whether or not the content can be transmitted by using, for example, the detection unit **201** every time a certain period of time elapses, and may generate updated determination information according to the detection result.

FIG. 5 is a diagram illustrating an example of a data structure of the determination information. The determination information in FIG. 5 is an example of determination information for determining whether four contents described in the delivery list in FIG. 4 can be transmitted. For example, "o" indicates that the content can be transmitted, and "x" indicates that the content cannot be transmitted. In the example of FIG. 5, four determination information ("o" or "x") corresponding to the four contents ("ContentA_1", "ContentA_2", "ContentA_3", and "ContentA_4") corresponding to four URLs of FIG. 4 are designated in the corresponding order.

If it is a method that can designate the determination information for each content, the determination information of the data structure other than FIG. 5 may be used. For example, the content identification information may be associated with the determination information. In order to be able to identify which delivery list the determination information corresponds to, the information for identifying the delivery list may be associated with the determination information.

That the content can be transmitted is for example, a state in which the content to be delivered is provided to the server **200** and can be transmitted to the client **300**. The content that cannot be transmitted means, for example, a state in which the content to be delivered has not yet been provided to the server **200** and cannot be transmitted to the client **300**. The server **200** may transmit the content (pull type transmission) in response to the request from the client **300**, or may transmit the content (push type transmission) without the request from the client **300**.

Regardless of whether the content is provided or not, the server **200** may designate whether or not the content can be transmitted. For example, in order to reduce the communication load, determination information indicating that transmission is not possible may be generated for a portion or all of a plurality of provided contents, and these contents may not be transmitted. In this way, the content that can be transmitted may not be transmittable thereafter.

The determination information may be one or may be plural for each content. For example, one or more metadata of the content may be used as the determination information. The metadata is, for example, range information indicating the range of the content, the data length of the content, the type of the content, and the like. The range information is, for example, information that designates the range of data to be delivered as the content among certain data. When the range information is confirmed, the confirmed range information is set as the determination information, and when the range information is unconfirmed, predetermined information (unconfirmed information) indicating that the range information is unconfirmed is set as the determination information. When the unconfirmed information is set in the

range information as the determination information, the client 300 can determine that the corresponding content cannot be transmitted. In this way, when the metadata is used as the determination information, it is sufficient to define a data format that can determine whether or not the content can be transmitted based on the metadata.

Returning to FIG. 3, the list transmission unit 213 transmits the delivery list generated by the list generation unit 211 to the client 300. For example, the list transmission unit 213 transmits the delivery list to the client 300 in advance before starting the transmission of the content. The determination information transmission unit 214 transmits the determination information generated by the determination information generation unit 212 to the client 300.

The request transmission/reception unit 215 transmits and receives various requests. For example, the request transmission/reception unit 215 receives a delivery list transmission request, a determination information transmission request, and a content transmission request from the client 300. In addition, the request transmission/reception unit 215 transmits a transmission request for the content stored in the storage unit 121 to the transmitting device 100.

The delivery unit 216 transmits the requested content to the client 300 that has transmitted the transmission request. When the push type transmission is adopted, the delivery unit 216 may transmit the content without the request from the client 300.

The data reception unit 217 receives the data transmitted from the transmitting device 100. For example, the data reception unit 217 receives the divided data as the data to be transmitted to the server 200. When a transmission request for data stored in the storage unit 121 is transmitted, the data reception unit 217 receives the data transmitted by the transmitting device 100 in response to the transmission request.

The storage control unit 218 controls storage processing for the temporary storage unit 221 and the storage unit 222. When the data transmitted from the transmitting device 100 and the data transmitted from the transmitting device 100 converted into a format to be delivered by the server 200 are configured to be stored in the storage unit 222 and then delivered, the processing of storing data in the storage unit 222 may not be able to keep up with a delivery speed, the processing of writing data to the storage unit 222 may freeze temporarily, or the like, resulting in an abnormality in the writing processing, which may prevent normal delivery. Therefore, the storage control unit 218 stores the data transmitted from the transmitting device 100 in the temporary storage unit 221 that is a storage medium that does not cause the above-mentioned abnormality in the writing processing. In the case where the data is stored in the temporary storage unit 221, the delivery unit 216 reads the data from the temporary storage unit 221 and delivers the data to the client 300. At this time, the data transmitted from the transmitting device 100 is finally stored in the storage unit 222. Therefore, the delivery unit 216 may operate as if it were delivered from the storage unit 222 in response to the request from the client 300. That is, the data saved in the temporary storage unit 221 may be returned in response to the request for the data saved in the storage unit 222. Such a function can be realized by recording the association of these files on the server 200 (database, file, symbolic link, etc.) and referencing it at the time of delivery.

Then, the storage control unit 218 performs processing (writing processing) of writing the data stored in the temporary storage unit 221 to the storage unit 222. In the writing processing, the storage control unit 218 may combine a

plurality of data stored in the temporary storage unit 221 into one data and store the combined data in the storage unit 222. For example, the storage control unit 218 may combine a plurality of data captured within a certain period of time so as to be included in one file, and write the combined file to the storage unit 222. As a result, it is possible to avoid a problem that the data cannot be stored because the number of files stored in the storage unit 222 exceeds the allowable amount of an operating system, for example.

After combining the plurality of data into one data and storing the combined data in the storage unit 222, when a transmission request for the data included in the combined data is received from the client 300, a function of specifying the corresponding data from the combined data is required. Therefore, for example, the list generation unit 211 may update the delivery list so as to include specifying information for specifying the corresponding data. The list generation unit 211 creates a delivery list in which the specifying information indicating a position of data in the combined file (byte offset from the beginning, etc.) is associated with the identification information (URL, etc.) of the data, and transmits the created delivery list to the client 300.

In the case where the client 300 requests the transmission of data by referring to the updated delivery list, the client 300 specifies the associated specifying information together with the identification information of the requested data. For example, the client 300 requests the data by adding the specifying information to a URL indicating the identification information of the data. The client 300 may request the data by the transmission request including the specifying information in a header (such as an HTTP extension header).

The file name requested at this time may be an original file name before division. This makes it easy to determine whether the data has already been received and cached on the client 300, and it is possible to suppress a re-acquisition of the acquired data. In this case, the server 200 needs to specify the combined data (combined file) from the original divided file (fragment file). For this purpose, there are methods in which (1) the information of the combined file for the fragment file is transmitted in advance as the specifying information to be transmitted to the client 300, and the client 300 describes it in the HTTP extension header at the time of request, and (2) the server 200 specifies the combined file by itself when responding to the request from the requested fragment file name. Regarding (2), it is conceivable to (2-1) record the association with the fragment file when creating the combined file (database, file, symbolic link, etc.), and (2-2) set the self-solving by the naming convention (for example, in the example of FIG. 4, the combined file name of ContentsA_1 to ContentsA_4 is ContentsA).

The server 200 specifies the corresponding data from the combined data using the specifying information transmitted from the client 300, and delivers the specified data to the requesting client 300.

The storage control unit 218 may delete the data stored in the storage unit 222 according to a predetermined condition. For example, the storage control unit 218 may delete the data for which a certain period of time has passed since the data was stored. The storage control unit 218 may delete the data step by step every time a certain period of time elapses. For example, the storage control unit 218 deletes all or a portion of the P picture from the storage unit 222 after a predetermined period (for example, one day) elapses, and thereafter, further deletes the P picture or the I picture from the storage unit 222 so as to thin out the P picture or the I picture step by step every time a predetermined period

11

(which may be the same as or different from the first period) elapses. By such processing, when it is desired to reduce a storage capacity for long-term recording, it is possible to obtain a video whose intermittence increases with the passage of time without performing, for example, processing of converting the image into a video with reduced quality. For the stepwise deletion of the P picture, for example, the encoding method described in Japanese Patent No. 6239472 can be used.

In addition, for example, the storage control unit **218** may preferentially delete the transmitted data. At this time, the storage control unit **218** may determine whether or not the data has been transmitted by using the metadata indicating whether or not the data has been transmitted. Further, the storage control unit **218** may prepare other metadata for reference in the deletion processing in advance. Such metadata is, for example, data that can determine which data corresponds to which picture. More specifically, the metadata includes, for example, the time (such as the date, time, minute, and second of the image), the position of the image within the time (for example, information indicating the number from the beginning), the file name, and a byte offset from the beginning of the file.

The temporary storage unit **221** temporarily stores the received data. For example, the temporary storage unit **221** can be configured by a volatile memory such as a Dynamic Random Access Memory (DRAM).

The storage unit **222** stores various data used by the server **200**. For example, the storage unit **222** stores the content to be delivered, the generated delivery list, and the generated determination information. The storage unit **222** can be configured by any commonly used storage medium such as a memory card, a Random Access Memory (RAM), a Hard Disk Drive (HDD), and an optical disk.

Note that if the above-mentioned abnormality in the writing processing does not occur, the temporary storage unit **221** may not be provided.

Each of the above units (the detection unit **201**, the list generation unit **211**, the determination information generation unit **212**, the list transmission unit **213**, the determination information transmission unit **214**, the request transmission/reception unit **215**, the delivery unit **216**, the data reception unit **217**, and the storage control unit **218**) is realized by, for example, one or a plurality of processors. For example, each of the above units may be realized by causing a processor such as a CPU to execute a program, that is, by software. Each of the above units may be realized by a processor such as a dedicated IC, that is, by hardware. Each of the units may be realized by using software and hardware in combination. When using a plurality of processors, each processor may realize one of the units or two or more of the units.

Note that each function of the server **200** may be delivered to a plurality of physically or logically different devices. For example, it may be divided into a server device that transmits a delivery list and a server device that transmits content. In addition, for example, it may be divided into a server device that receives data from the transmitting device **100** and stores the data in the temporary storage unit **221** and the storage unit **222**, and a server device that reads data from the temporary storage unit **221** and the storage unit **222** and delivers the data.

Next, a function of the client **300** will be described. As illustrated in FIG. 3, the client **300** includes a list reception unit **311**, a determination information reception unit **312**, a

12

determination unit **313**, a request transmission unit **314**, a data reception unit **315**, a reproduction unit **316**, and a storage unit **321**.

The list reception unit **311** receives the delivery list from the server **200**. The determination information reception unit **312** receives the determination information from the server **200**.

The determination unit **313** determines the content for which transmission is requested based on the delivery list and the determination information. For example, the determination unit **313** determines the content in which "o" is set as the determination information as illustrated in FIG. 5 among the contents whose URL is described in the delivery list illustrated in FIG. 4, as the content for which transmission is requested. In the case where the above-mentioned range information is used as the determination information, the determination unit **313** determines, for example, the content in which the unconfirmed information is not set in the range information as the content for which transmission is requested. In the case where a plurality of determination information is used, the determination unit **313** may determine the content for which transmission is requested according to a combination of the plurality of determination information. For example, in the case where all the determination information indicates that the content can be transmitted, the determination unit **313** determines that the corresponding content is the content for which transmission is requested.

The request transmission unit **314** transmits a transmission request for the content determined to request transmission to the server **200**. The data reception unit **315** receives the content transmitted in response to the transmission request transmitted by the request transmission unit **314** from the server **200**. The reproduction unit **316** reproduces the received content.

The storage unit **321** stores various data used by the client **300**. For example, the storage unit **321** stores the transmitted delivery list, the transmitted determination information, and the delivered content.

Each of the above units (the list reception unit **311**, the determination information reception unit **312**, the determination unit **313**, the request transmission unit **314**, the data reception unit **315**, and the reproduction unit **316**) is realized by, for example, one or a plurality of processors. For example, each of the above units may be realized by causing a processor such as a CPU to execute a program, that is, by software. Each of the above units may be realized by a processor such as a dedicated IC, that is, by hardware. Each of the units may be realized by using software and hardware in combination. When using a plurality of processors, each processor may realize one of the units or two or more of the units.

By using the delivery list and the determination information as described above, the server **200** does not need to create and transmit the delivery list every time there is an update, for example. In addition, the client **300** can more easily analyze the updated part by referring to the determination information. That is, it is possible to efficiently acquire and analyze the same information as the latest delivery list without reacquiring and reanalyzing the delivery list.

Note that, without using the determination information, for example, the delivery list may be created and transmitted every time the data is updated. In this case, the functions related to the determination information (the determination information generation unit **212**, the determination information transmission unit **214**, the determination information

13

reception unit 312, etc.) may not be provided. In addition, when, for example, the push type transmission is adopted, the server 200 may deliver the video to the client 300 without using the delivery list.

Next, data transmission processing by the transmitting device 100 according to the present embodiment will be described. FIG. 6 is a flowchart illustrating an example of transmission processing in the present embodiment.

The imaging unit 101 captures a video to be delivered (step S101). The encoding unit 111 encodes the video input from the imaging unit 101 (step S102). The division unit 112 divides the data to be transmitted to the server 200 and the data to be stored in the storage unit 121. For example, the division unit 112 divides the input video into a fragment of an I picture and a fragment of a P picture (step S103). The data transmission unit 113 transmits the fragment of the I picture to the server 200 (step S104). The storage control unit 114 stores the fragment of the P picture in the storage unit 121 (step S105).

Next, transmission processing of the data stored in the storage unit 121 will be described. FIG. 7 is a flowchart illustrating an example of transmission processing of the data stored in the storage unit 121.

The request reception unit 115 receives a transmission request for data stored in the storage unit 121 from the server 200 (or the control server) (step S201). In the case where the P picture is stored in the storage unit 121, the request reception unit 115 receives the transmission request of any P picture among the stored P pictures. The data transmission unit 113 reads the requested P picture from the storage unit 121 and transmits the P picture to the server 200 (step S202).

Next, list generation processing by the server 200 according to the present embodiment configured in this way will be described. The list generation processing is processing in which the server 200 generates the delivery list. FIG. 8 is a flowchart illustrating an example of the list generation processing in the present embodiment.

The list generation unit 211 of the server 200 generates the delivery list in response to a request from, for example, the client 300 (step S301). The list generation unit 211 stores the generated delivery list in, for example, the storage unit 222 (step S302).

As an example, it is assumed that ContentA_1, ContentA_2, ContentA_3, and ContentA_4, which are temporally continuous contents, exist as contents to be delivered, and ContentA_1 and ContentA_3 can be delivered. In this case, the list generation unit 211 generates a delivery list for delivering ContentA_1, ContentA_2, ContentA_3, and ContentA_4. FIG. 4 described above illustrates an example of the delivery list generated at this time.

Next, determination information generation processing by the server 200 according to the present embodiment will be described. The determination information generation processing is processing in which the server 200 generates the determination information. The determination information generation processing is executed, for example, when the client 300 requests transmission of determination information. FIG. 9 is a flowchart illustrating an example of the determination information generation processing in the present embodiment.

The determination information generation unit 212 of the server 200 determines whether or not the determination information has been generated for the content to be delivered (step S401). If the determination information has not been generated (step S401: No), the determination information generation unit 212 generates the determination information for the content (step S402). FIG. 5 described above

14

illustrates an example of the determination information generated for the example described in FIG. 8 (an example in which ContentA_1 and ContentA_3 can be delivered).

After the determination information is generated, and when the determination information has already been generated (step S401: Yes), the determination information generation unit 212 determines whether or not to update the generated determination information (step S403). For example, the determination information generation unit 212 determines that the determination information is updated when a detection result indicating that the content can be transmitted is received from the detection unit 201.

If it is determined that the determination information is to be updated (step S403: Yes), the determination information generation unit 212 updates the determination information (step S404). After updating the determination information, and when it is determined that the determination information is not updated (step S403: No), the determination information generation processing is terminated.

The server 200 makes the generated delivery list and determination information available for delivery via the network 402. The client 300 can access the server 200 to acquire the delivery list and the determination information.

Next, content delivery processing by the communication system according to the present embodiment will be described. FIG. 10 is a sequence diagram illustrating an example of content delivery processing in the present embodiment.

The list generation unit 211 of the server 200 generates the delivery list (step S501). Such processing corresponds to, for example, the above-mentioned list generation processing. The list transmission unit 213 of the server 200 transmits the delivery list to the client 300, for example, in response to the request from the client 300 (step S502).

The determination information generation unit 212 of the server 200 generates the determination information (step S503). Such processing corresponds to, for example, the above-mentioned determination information generation processing. The determination information transmission unit 214 of the server 200 transmits the determination information to the client 300, for example, in response to the request from the client 300 (step S504).

The list reception unit 311 of the client 300 receives the delivery list, and the determination information reception unit 312 receives the determination information. After that, the determination unit 313 of the client 300 determines the content for which transmission is to be requested by using the received delivery list and the determination information (step S505). For example, in the case where the delivery list illustrated in FIG. 4 is received, the determination unit 313 obtains URLs for acquiring ContentA_1 to ContentA_4 by analyzing the received delivery list. In addition, the determination unit 313 analyzes the received determination information and determines the content that can be transmitted among the contents described in the delivery list.

FIG. 11 is a diagram illustrating an example of determining the delivery list illustrated in FIG. 5 by using the determination information illustrated in FIG. 4. As illustrated in FIG. 11, by combining the URL for acquiring ContentA_1 described in the delivery list and the determination information corresponding to ContentA_1, the determination unit 313 can determine that ContentA_1 can be transmitted (accessible). Similarly, the determination unit 313 can determine that ContentA_2 cannot be transmitted (inaccessible).

Returning to FIG. 10, the request transmission unit 314 transmits a transmission request for the content determined

15

to request transmission to the server **200** (step **S506**). The delivery unit **216** of the server **200** transmits the requested content to the client **300** (step **S507**). The data reception unit **315** of the client **300** receives the content, and the reproduction unit **316** reproduces the received content (step **S508**).

In the case where the content expressed in a format that can be delivered as it is by the server **200** is transmitted from the transmitting device **100**, the delivery unit **216** does not convert the content transmitted from the transmitting device **100** and directly transmits the content to the client **300**. In the case where the content is transmitted in a format to which metadata that can be converted into a format delivered by the server **200** is added, the delivery unit **216** converts the transmitted content into a format that can be delivered according to the metadata, and transmits the converted content to the client **300**.

Thereafter, it is assumed that ContentA_2 is ready for delivery. At this time, the detection unit **201** of the server **200** detects that ContentA_2 is ready for delivery. The determination information generation unit **212** generates the determination information updated according to the detection result (step **S509**).

FIG. **12** is a diagram illustrating an example of the updated determination information. As illustrated in FIG. **12**, the determination information generation unit **212** updates the determination information corresponding to ContentA_2 from "x" to "o". The updated determination information is made ready for delivery via the network **402**.

Returning to FIG. **10**, the determination information transmission unit **214** of the server **200** transmits the updated determination information to the client **300** in response to the request from the client **300**, for example (step **S510**). Subsequent steps **S511** to **S514** are the same as steps **S505** to **S508**.

FIG. **13** is a diagram illustrating an example of determining the delivery list illustrated in FIG. **4** by using the determination information illustrated in FIG. **12**. As illustrated in FIG. **13**, the determination unit **313** can determine that ContentA_2 is in a transmittable (accessible) state.

In this way, when ContentA_2 is ready for delivery, the client **300** does not reacquire the delivery list, but can obtain the latest state of each content only by simply applying the updated determination information to the delivery list that has already been received.

The present embodiment can be applied to, for example, a system for monitoring a video captured by a drive recorder. Applicable systems are not limited thereto. For example, it may be applied to a system that delivers and monitors sensor data obtained by a sensor, and a system that delivers and monitors moving image data obtained by an imaging device mounted on a moving body or the like. The moving body includes, for example, people, robots, vehicles (automobiles, motorcycles, trains, etc.), trolleys, flyable objects (manned aerial vehicles, unmanned aerial vehicles (e.g., UAVs), drones), and personal mobility, and so on. In addition, the moving body is, for example, a moving body that travels through a driving operation by a person, and a moving body that can automatically travel (automatically drive) without a driving operation by a person.

The present embodiment can also be applied to, for example, a monitoring system that monitors an operation history of a screen (Human Machine Interface (HMI)) of a monitoring and control system of the device. The operation history of the screen can be obtained by, for example, a function of capturing and recording the screen displayed on

16

the display device. The image data thus obtained can be used in place of the video captured by the transmitting device **100**.

Since a large number of monitoring screens are usually used in the monitoring and control system of the device, it may be necessary for the monitoring system of the operation history of the screen to monitor a large number of monitoring screen images side by side. In general, the processing of displaying a large number of videos in parallel in this way increases a processing load, but in the present embodiment, since only the divided data (for example, I picture) can be displayed, the increase in the processing load can be suppressed.

Modified Example 1

The network **401** may be configured by a plurality of communication lines, and may be configured to transmit a portion (for example, I picture) of the data divided by the division unit **112** to the server **200** by one of the plurality of communication lines (communication line **401A**), and transmit the remaining data (for example, P picture) to the server **200** by the other communication line (communication line **401B**) among the plurality of communication lines. FIG. **14** is a diagram illustrating a configuration example of a video delivery system of the modified example 1 configured in this way. For example, the data transmission unit **113** may live-stream the I picture by the communication line **401A** and transmit the P picture by the communication line **401B** at any timing.

Modified Example 2

Instead of transmitting the divided data to the server **200**, the divided data may be divided and stored in the storage unit **121** in the transmitting device **100**. As a result, for example, it is possible to read and display only a necessary picture (for example, an I picture) from the storage unit **121**. The method of displaying only the I picture can reduce the processing load as compared with the method of displaying all the pictures including the P picture. Therefore, for example, even for the client **300** or the like whose processing capacity is limited, it is possible to read only the I pictures from the storage units **121** of a plurality of transmitting devices **100** and display them in parallel. Further, if necessary, the remaining P pictures and the like can be read step by step to make a moving display.

In this case, the storage control unit **114** may delete the data for which a certain period of time has passed since it was stored, as in the case of the storage control unit **218** of the server **200**, and may delete the data step by step after a certain period of time has elapsed. For example, when the storage capacity is insufficient, it is possible to delete only the P picture and leave the I picture as data that can be displayed intermittently, instead of deleting all the data including the I picture and the P picture in chronological order, and therefore, both long-term storage of data and low capacity can be achieved.

Modified Example 3

If the video is divided and delivered for each picture, a size of the delivery list may increase. Therefore, the list generation unit **211** may create a delivery list from a plurality of files having a hierarchical structure. For example, the list generation unit **211** may configure the delivery list hierarchically in units of year, month, day, time,

17

and the like. For example, the delivery list at the bottom layer includes identification information of each piece of fragment data divided into a certain range of time. The next top delivery list includes information that identifies one or more delivery lists that correspond to the time included in a day. The next top delivery list includes information that identifies one or more delivery lists that correspond to the days included in a month. The next top delivery list includes information that identifies one or more delivery lists that correspond to the months included in a year.

Modified Example 4

FIG. 15 is a diagram illustrating an outline of delivery processing by a video delivery system of the modified example 4. The video delivery system of the present modified example further includes a control server 510 and a Web server 520. In the present modified example, the server 200 is configured as a Web server that delivers video.

The control server 510 includes a control unit 511. The control unit 511 controls the transmitting device 100 according to the request transmitted from the client 300. For example, the client 300 transmits a transmission request to the control server 510 indicating whether to divide the picture and transmit only a portion (for example, only the I picture) or all the pictures. The control unit 511 of the control server 510 transmits, according to the transmission request, a control signal indicating whether to divide only a portion of the pictures and transmit (upload) them to the server 200, or transmit all the pictures to the server 200, to the transmitting device 100.

The transmitting device 100 operates by switching between transmitting of only a portion of the pictures divided as in the above embodiment to the server 200 or transmitting all the pictures to the server 200, according to the control signal transmitted from the control server 510.

The capture stream transmission (upload transmission) means that the divided data is transmitted to the server 200 as the data (first data) to be transmitted to the server 200. The capture file save (file output) means that the divided data is stored as the data (second data) to be stored in the storage unit 121 without being transmitted to the server 200. As illustrated in FIG. 15, the stored data may be copied offline to the server 200.

The server 200 converts the data received by the data reception unit 217 into a deliverable format (stream conversion) and stores it in the storage unit 222, if necessary. The data copied offline may be collectively converted (batch conversion) and stored in the storage unit 222.

The storage unit 222 stores each picture (fragment file) in a deliverable format, a delivery list, and the like. The server 200 functions as a video server that delivers the pictures stored in the storage unit 222. As described above, the data received by the data reception unit 217 may be stored in the temporary storage unit 221 and the data stored in the temporary storage unit 221 may be delivered by the delivery unit 216.

The client 300 has a web browser with a viewer. The web browser has a video player that reproduces a video. The video player is realized as, for example, an HTML5-compliant application.

The Web server 520 is a server device having a function other than the function of the video server. For example, the Web server 520 is a server device for providing a user interface (UI) for an external system different from the video delivery system.

18

As described above, the video delivery system of the above embodiment can be realized as a Web system used on the Internet, for example.

Modified Example 5

The data deleted from the storage unit 222 may be stored (saved, moved) in another storage medium. FIG. 16 is a block diagram illustrating an example of a functional configuration of each device of the video delivery system of the modified example 5.

As illustrated in FIG. 16, the video delivery system of the present modified example further includes a storage device 600-2. In addition, a function of a storage control unit 218-2 of a server 200-2 is different from that of the storage control unit 218 of the above embodiment. Since other configurations and functions are the same as those in FIG. 3, which is a block diagram of the video delivery system according to the above embodiment, the same reference numerals are given and the description thereof is omitted herein.

The storage device 600-2 (an example of the second storage unit) is a device including a storage medium for storing data moved from the storage unit 222 (an example of the first storage unit) of the server 200-2. The storage device 600-2 may be, for example, a storage device constructed on a cloud environment. The storage medium can be configured by any commonly used storage medium such as a memory card, a RAM, a HDD, an optical disk, and a tape.

The storage device 600-2 may be configured by a storage medium having a lower cost than the storage unit 222. By using the storage device 600-2 including such a storage medium, the cost for data storage can be reduced even when the amount of data to be stored is increased and a data retention period is extended.

The storage device 600-2 is connected to the server 200-2 via a network such as the Internet. Such a network may be the same as at least one of the networks 401 and 402, or may be a different network.

The storage device 600-2 may be provided in the server 200-2. For example, the storage device 600-2 as a storage unit (an example of the second storage unit) including a storage medium different from the storage unit 222 may be provided in the server 200-2.

In addition to the functions of the storage control unit 218, the storage control unit 218-2 includes a function of deleting data stored in the storage unit 222 according to predetermined conditions from the storage unit 222 and storing data including at least a portion of the deleted data in the storage device 600-2.

The data including at least a portion of the deleted data means that the data may be not only all of the deleted data, only a portion of the deleted data, or only the deleted data (portion or all), but also include data that has not been deleted (data that is also stored in the storage unit 222).

FIG. 17 is a diagram illustrating an example of how to move data. FIG. 17 illustrates an example in which data is moved step by step every time a certain period of time elapses. $t_0, t_1, \dots, t_N$ represent the time (seconds) attached to the data. In addition, FIG. 17 illustrates an example in which data is stored at 60 frames per second (I picture: two frames, P picture: 58 frames).

In the first stage, the storage control unit 218-2 deletes P pictures 1701a and 1701b among the data stored in the storage unit 222, and stores the deleted P pictures 1701a and 1701b in the storage device 600-2. As a result, data of two frames per second (I picture only) is left in the storage unit 222.

In the second stage, the storage control unit **218-2** deletes an I picture **1702** from the data of two frames per second stored in the storage unit **222**, and stores the deleted I picture **1702** in the storage device **600-2**. As a result, data of one frame per second (I picture only) is left in the storage unit **222**.

In the third stage, the storage control unit **218-2** deletes an I picture **1703**, which is the data of one frame per second stored in the storage unit **222**, at regular intervals, and stores the deleted I picture in the storage device **600-2**. For example, the storage control unit **218-2** halves the data (I picture) in the I picture **1703** so that it becomes one frame in two seconds.

Note that the movement method illustrated in FIG. **17** is an example and is not limited thereto. For example, the condition for moving data is not limited to only the elapse of a certain period of time as illustrated in FIG. **17**, but may be other conditions. In addition, the method of determining the data to be moved is not limited to the method illustrated in FIG. **17**. In FIG. **17**, an example is illustrated in which the P picture is moved first and then the I picture is moved step by step. The storage control unit **218-2** may move the data in units of a group of pictures (GOP) including the I picture and the P picture. In the example of FIG. **17**, the storage control unit **218-2** may move one of the two GOPs included in one second.

When a predetermined condition (storage condition) is satisfied, the storage control unit **218-2** may control the data so that the data is not stored in the storage unit **222** but stored in the storage device **600-2** from the beginning. For example, the storage control unit **218-2** stores data determined to be unimportant in the storage device **600-2**, and stores data determined to be important in the storage unit **222**. Thereafter, when yet another condition (condition for moving data) is satisfied, the storage control unit **218-2** deletes the data stored in the storage unit **222** from the storage unit **222** and stores the data in the storage device **600-2**.

Any method may be used to determine whether or not the data is important, but for example, a method of making determination using information input from the outside together with the data and a method of making determination using the importance (priority) given to the data can be applied.

The information input from the outside is, for example, sensor information of a device (vehicle or the like) on which the transmitting device **100** is mounted. When the device on which the transmitting device **100** is mounted is a vehicle, the sensor information includes, for example, information indicating acceleration/deceleration, information indicating whether sudden braking has been applied, information indicating a sudden change in accelerator opening, information indicating steering such as a sudden steering wheel, and the like.

For example, the storage control unit **218-2** determines whether or not the sudden acceleration/deceleration has occurred from the information indicating acceleration/deceleration, and when it is determined that the sudden acceleration/deceleration has occurred, the storage control unit **218-2** stores the data transmitted from the corresponding transmitting device **100** in the storage unit **222**. The storage control unit **218-2** stores the data transmitted from the corresponding transmitting device **100** in the storage device **600-2** when it is determined that the sudden acceleration/deceleration has not occurred.

By such processing, it becomes possible to control so that more important data is stored in the storage unit **222** and

other data is stored in the storage device **600-2**. By configuring the storage unit **222** to store only more important data, the storage capacity required for the storage unit **222** can be reduced.

As described above, in the video delivery system according to the present embodiment, it is possible to reduce the line load and the processing load without deteriorating the quality.

Next, a hardware configuration of the transmitting device according to the present embodiment will be described with reference to FIG. **18**. FIG. **18** is an explanatory diagram illustrating a hardware configuration example of the transmitting device according to the present embodiment.

The transmitting device according to the present embodiment is a control device such as a CPU **51**, a storage device such as a Read Only Memory (ROM) **52** or a RAM **53**, a communication I/F **54** that connects to a network for communication, and a bus **61** that connects each unit.

A program executed by the transmitting device according to the present embodiment is provided by being incorporated in the ROM **52** or the like in advance.

The program executed by the transmitting device according to the present embodiment may be configured to be recorded on a computer-readable recording medium such as Compact Disk Read Only Memory (CD-ROM), flexible disks (FD), Compact Disk Recordable (CD-R), and Digital Versatile Disk (DVD) in a file in an installable format or an executable format and provided as a computer program product.

Further, the program executed by the transmitting device according to the present embodiment may be stored on a computer connected to a network such as the Internet and provided by downloading via the network. In addition, the program executed by the transmitting device according to the present embodiment may be configured to be provided or delivered via a network such as the Internet.

The program executed by the transmitting device according to the present embodiment can make the computer function as each unit of the transmitting device described above. Such a computer can read and execute a program on a main storage device from a computer-readable storage medium by the CPU **51**.

While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiment described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiment described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

What is claimed is:

1. A server device comprising:

one or more processors configured to:

receive first data that is a portion of a plurality of transmission data, from a transmitting device,
receive second data, other than the first data, among the plurality of transmission data, the second data being transmitted from the transmitting device in response to a transmission request for the second data,
control storing, in first storage, the first data, the second data transmitted from the transmitting device, and metadata indicating that the second data is not transmitted to a receiving device, the metadata being data that differs from one or more pieces of determination

21

information, the one or more pieces of determination information being transmitted to the receiving device, each of the one or more pieces of determination information being used for determining whether or not to request transmission of the transmission data, and

delete, based on the metadata, at least one of the first data and the second data stored in the first storage from the first storage according to a predetermined condition, and store data including at least a portion of the deleted data in second storage.

2. The device according to claim 1, wherein the transmission data is moving image data, and the first data includes image data encoded by intra-frame prediction.

3. The device according to claim 2, wherein the second data includes image data encoded by inter-frame prediction based on the first data.

4. The device according to claim 1, further comprising temporary storage configured to store the first data and the second data received, wherein the one or more processors:

store, in the first storage, the first data and the second data stored in the temporary storage, and

deliver the first data stored in the temporary storage to the receiving device.

5. The device according to claim 4, wherein the one or more processors store, in the first storage, combined data in which at least portions of the first data and the second data stored in the temporary storage are combined.

6. A server device comprising:

one or more processors configured to:

deliver first data to a receiving device, the first data being a portion of a plurality of transmission data, the first data being transmitted from a transmitting device, and the first data being stored in first storage,

deliver second data to the receiving device, the second data being data other than the first data among the plurality of transmission data, the second data being transmitted from a transmitting device in response to a transmission request for the second data, and the second data being stored in the first storage, and

delete, based on metadata, at least one of the first data and the second data stored in the first storage from the first storage according to a predetermined condition, and store data including at least a portion of the deleted data in second storage, the metadata indicating that the second data is not transmitted to the receiving device, the metadata being data that differs from one or more pieces of determination information, the one or more pieces of determination information being transmitted to the receiving device, each of the one or more pieces of determination information being used for determining whether or not to request transmission of the transmission data.

7. The device according to claim 6, wherein when a predetermined storage condition is satisfied, the one or more processors do not store the first data and the second data in the first storage, but store the first data and the second data in the second storage.

8. The device according to claim 6, wherein the transmission data is moving image data, and the first data includes image data encoded by intra-frame prediction.

22

9. The device according to claim 8, wherein the second data includes image data encoded by inter-frame prediction based on the first data.

10. The device according to claim 6, wherein the one or more processors are configured to:

receive a transmission request for the first data or the second data from the receiving device, the transmission request including specifying information for specifying a position of the first data or the second data in combined data, the combined data being stored in the first storage and the combined data being data in which at least portions of the first data and the second data stored in temporary storage are combined, and deliver the first data or the second data read from the first storage to the receiving device based on the specifying information.

11. An information processing method comprising:

receiving first data that is a portion of a plurality of transmission data, from a transmitting device;

receiving second data, other than the first data, among the plurality of transmission data, the second data being transmitted from the transmitting device in response to a transmission request for the second data;

controlling storing, in first storage, the first data, the second data transmitted from the transmitting device, and metadata indicating that the second data is not transmitted to a receiving device, the metadata being data that differs from one or more pieces of determination information, the one or more pieces of determination information being transmitted to the receiving device, each of the one or more pieces of determination information being used for determining whether or not to request transmission of the transmission data; and deleting, based on the metadata, at least one of the first data and the second data stored in the first storage from the first storage according to a predetermined condition, and store data including at least a portion of the deleted data in second storage.

12. A computer program product comprising a non-transitory computer-readable medium including programmed instructions, the instructions causing a computer to execute:

receiving first data that is a portion of a plurality of transmission data, from a transmitting device;

receiving second data, other than the first data, among the plurality of transmission data, the second data being transmitted from the transmitting device in response to a transmission request for the second data;

controlling storing, in first storage, the first data, the second data transmitted from the transmitting device, and metadata indicating that the second data is not transmitted to a receiving device, the metadata being data that differs from one or more pieces of determination information, the one or more pieces of determination information being transmitted to the receiving device, each of the one or more pieces of determination information being used for determining whether or not to request transmission of the transmission data; and deleting, based on the metadata, at least one of the first data and the second data stored in the first storage from the first storage according to a predetermined condition, and store data including at least a portion of the deleted data in second storage.

13. An information processing method comprising:

delivering first data to a receiving device, the first data being a portion of a plurality of transmission data, the

23

first data being transmitted from a transmitting device, and the first data being stored in first storage; delivering second data to the receiving device, the second data being data other than the first data among the plurality of transmission data, the second data being transmitted from a transmitting device in response to a transmission request for the second data, and the second data being stored in the first storage; and deleting, based on metadata, at least one of the first data and the second data stored in the first storage from the first storage according to a predetermined condition, and storing, in second storage, data including at least a portion of the deleted data, the metadata indicating that the second data is not transmitted to the receiving device, the metadata being data that differs from one or more pieces of determination information, the one or more pieces of determination information being transmitted to the receiving device, each of the one or more pieces of determination information being used for determining whether or not to request transmission of the transmission data.

14. A computer program product comprising a non-transitory computer-readable medium including programmed instructions, the instructions causing a computer to execute:

24

delivering first data to a receiving device, the first data being a portion of a plurality of transmission data, the first data being transmitted from a transmitting device, and the first data being stored in first storage; delivering second data to the receiving device, the second data being data other than the first data among the plurality of transmission data, the second data being transmitted from a transmitting device in response to a transmission request for the second data, and the second data being stored in the first storage; and deleting, based on metadata, at least one of the first data and the second data stored in the first storage from the first storage according to a predetermined condition, and storing, in second storage, data including at least a portion of the deleted data, the metadata indicating that the second data is not transmitted to the receiving device, the metadata being data that differs from one or more pieces of determination information, the one or more pieces of determination information being transmitted to the receiving device, each of the one or more pieces of determination information being used for determining whether or not to request transmission of the transmission data.

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